

Evaluating a Production RAG System

Retrieval quality should be measured before and after introducing changes such as hybrid search or cross-encoder reranking. Reproducible evaluation helps determine whether a change genuinely improves the system.

Core Retrieval Metrics

Metric	Description
Top-1 Accuracy	Percentage of queries where a relevant document ranks first.
Top-k Accuracy	Percentage of queries where a relevant document appears within the first k results.
Candidate Recall	Whether retrieval finds relevant evidence before reranking.
Preferred-Source Accuracy	Whether the most authoritative document receives the highest rank.

Evaluation Workflow

- 1 Create questions with known relevant documents.
- 2 Run vector, BM25, and hybrid retrieval independently.
- 3 Record Top-1 and Top-k accuracy.
- 4 Apply cross-encoder reranking to the same candidate set.
- 5 Compare reranked results with the original baseline.
- 6 Retain changes only when measured quality improves.

Production Guidance

Evaluation datasets should contain realistic queries, multiple acceptable sources where appropriate, and preferred-source labels when one document is more authoritative than another. Retrieval accuracy, latency, grounding, citation correctness, and generation quality should be measured separately.

A cross-encoder typically improves final ordering by scoring the question and candidate chunk together. However, it adds latency, so quality gains should be validated against performance costs.